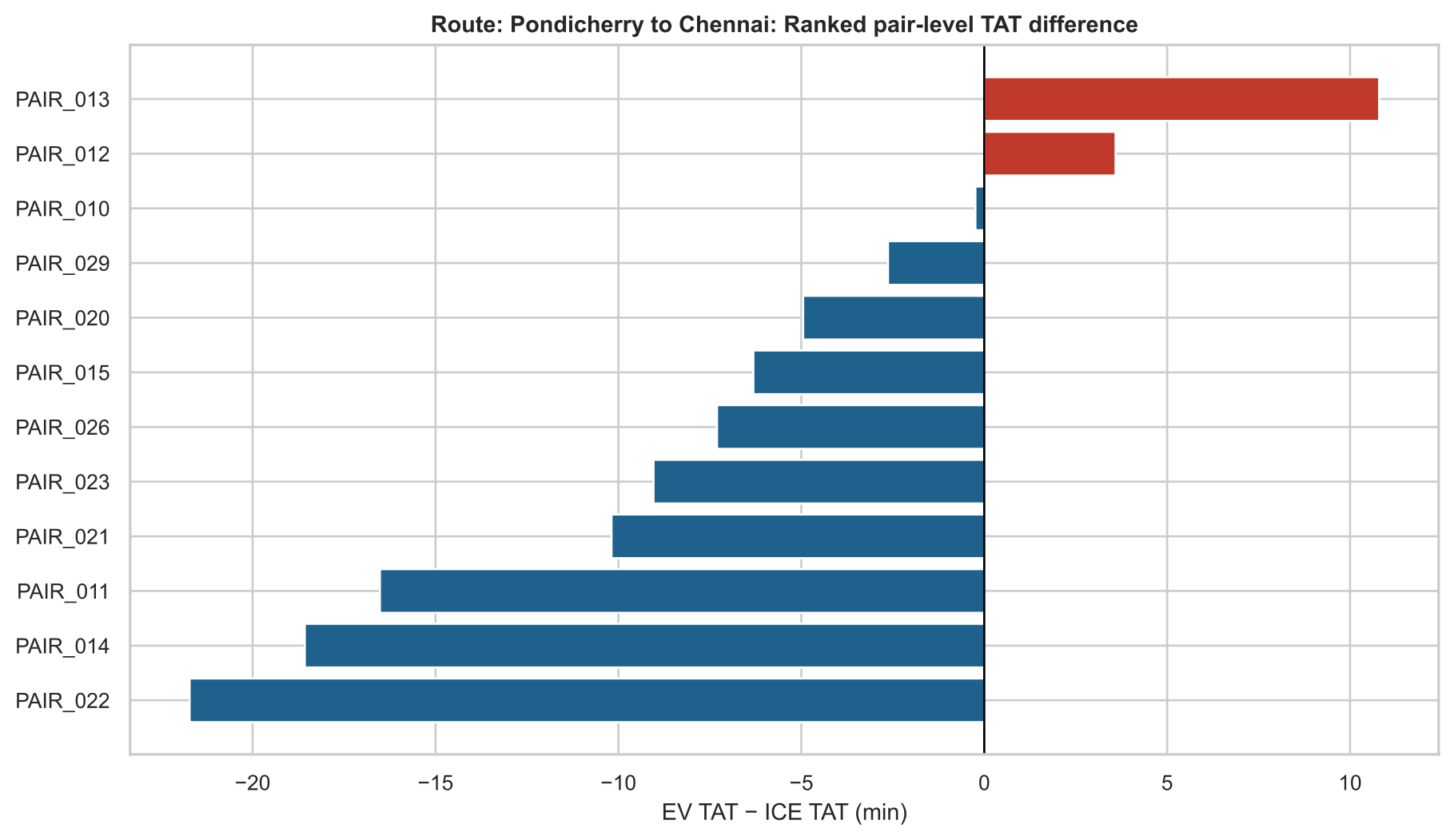
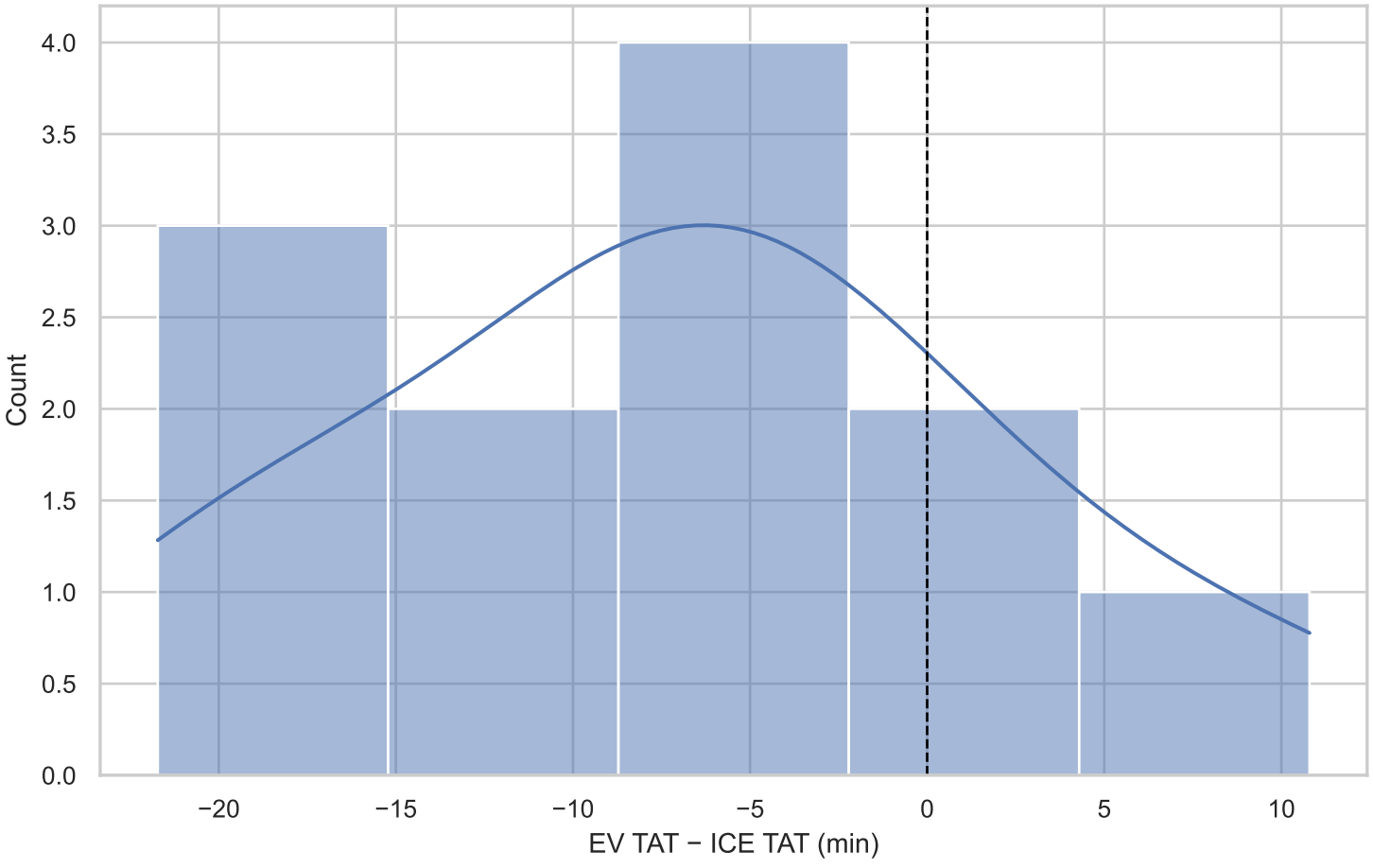
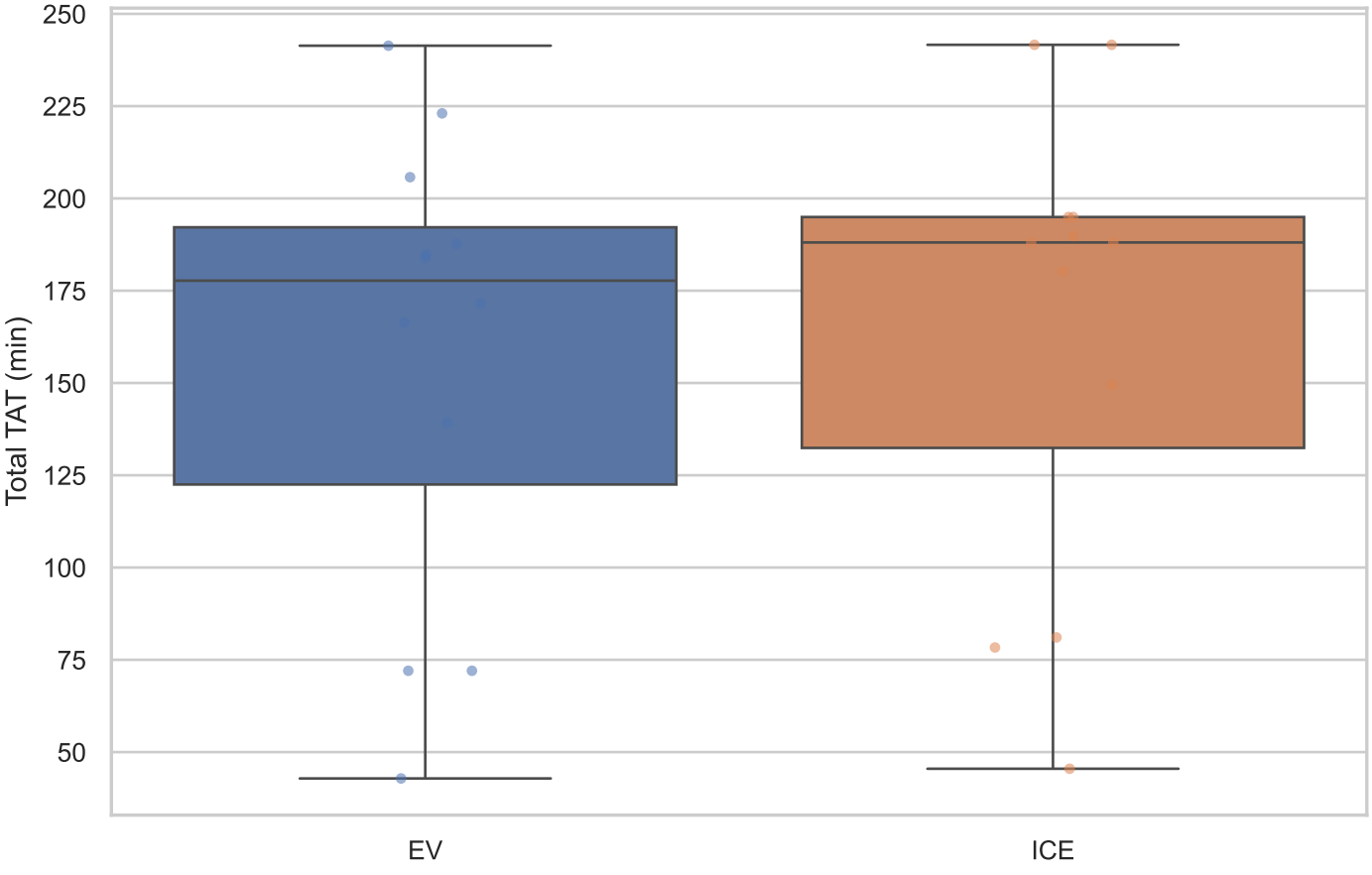
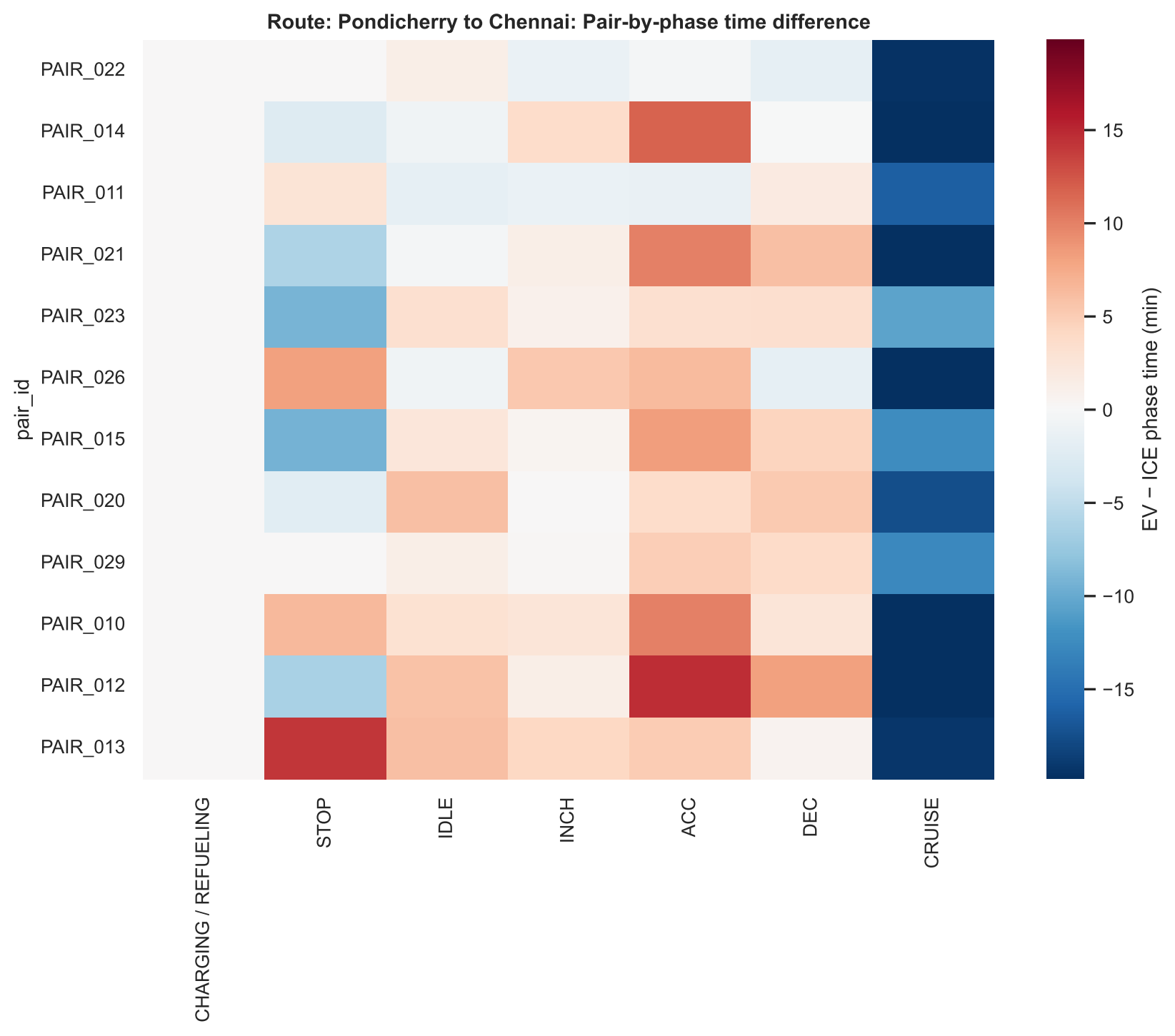
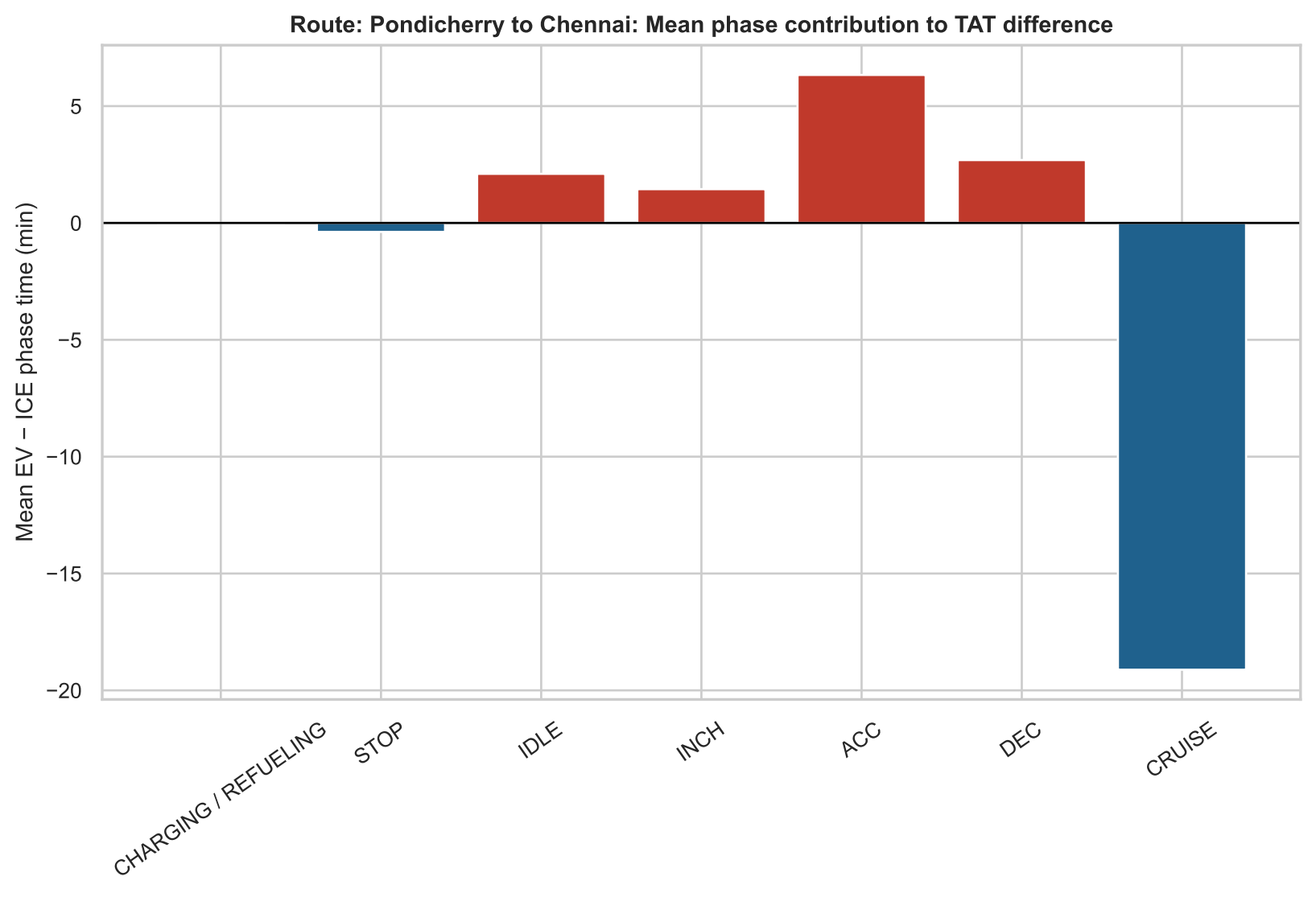
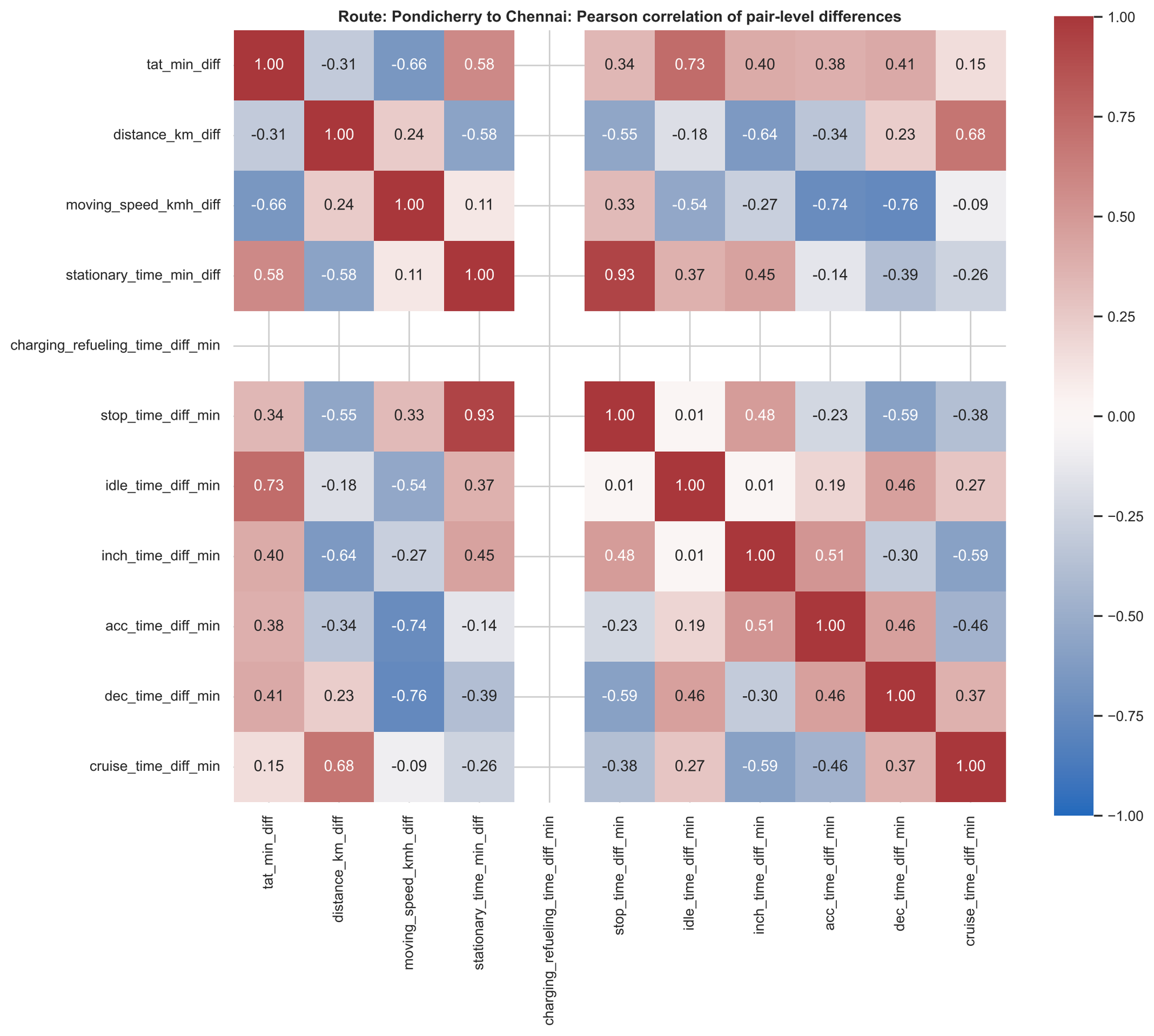

Route: Pondicherry to Chennai: Ranked pair-level TAT difference
PAIR_013
PAIR_012
PAIR_010
PAIR_029
PAIR_020
PAIR_015
PAIR_026
PAIR_023
PAIR_021
PAIR_011
PAIR_014
PAIR_022
−20
−15
−10
−5
0
5
10
EV TAT − ICE TAT (min)

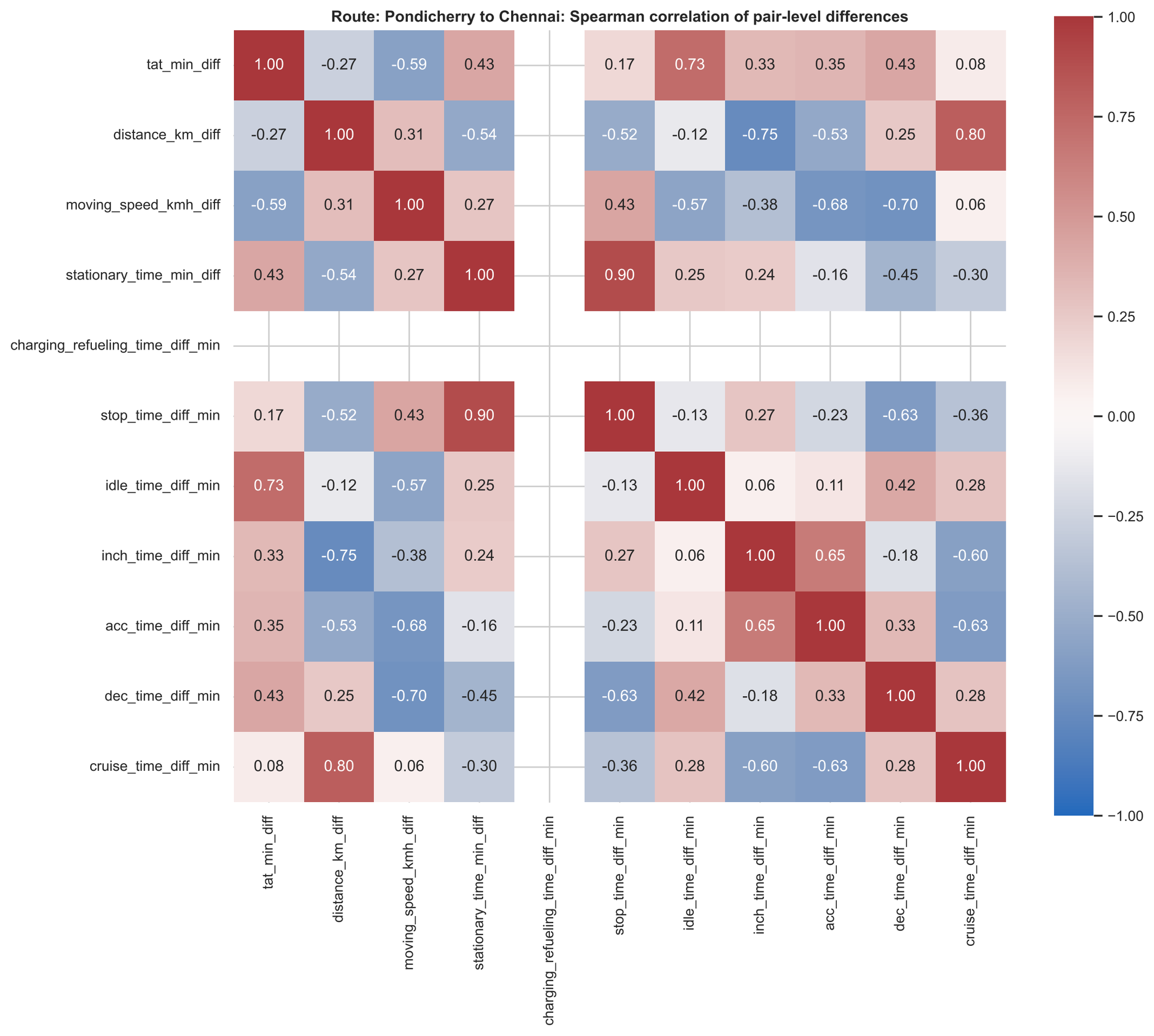

Route: Pondicherry to Chennai: Distribution of TAT differences
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−20
−15
−10
−5
0
5
10
EV TAT − ICE TAT (min)

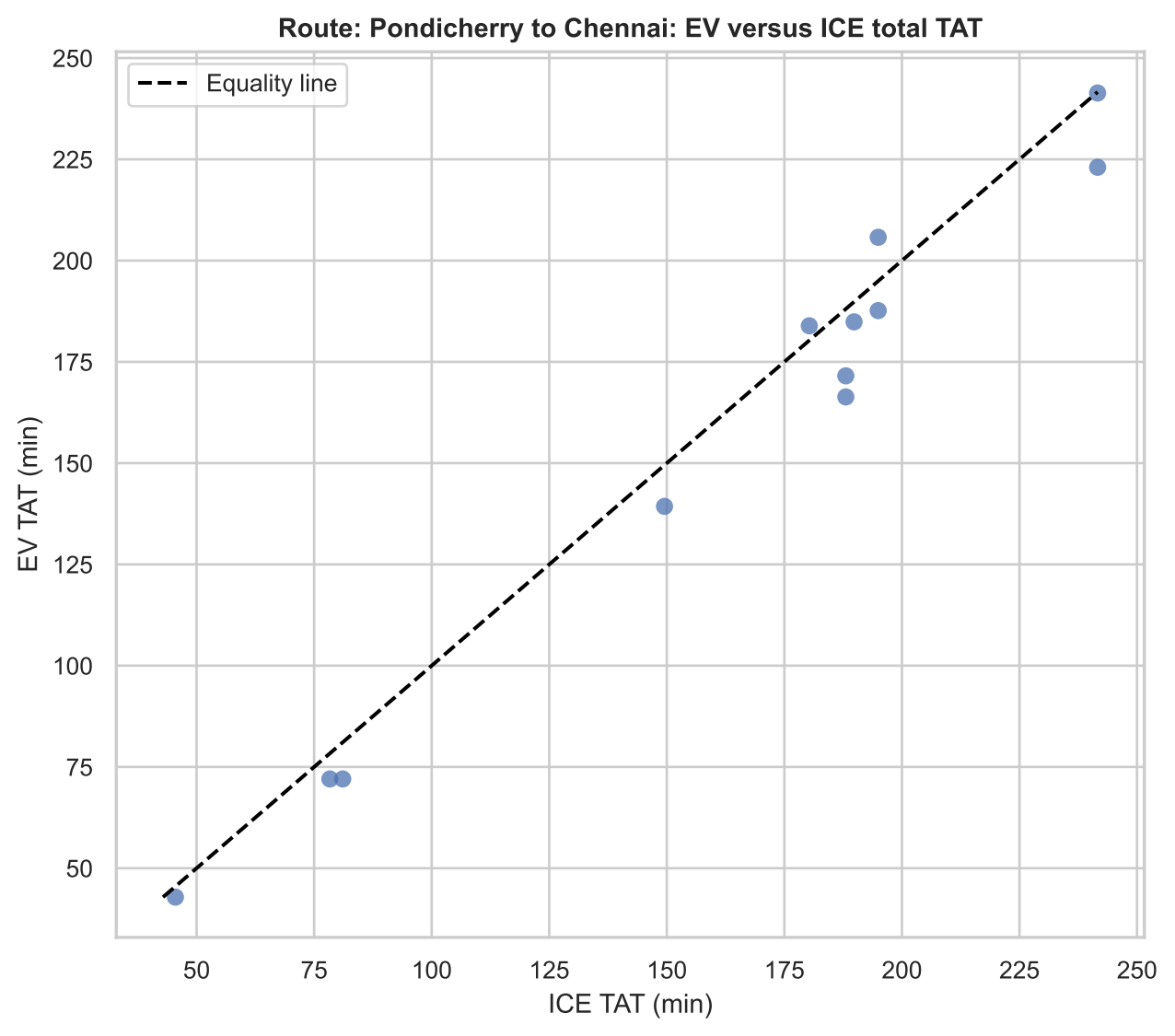

Route: Pondicherry to Chennai: EV and ICE TAT distribution
Total TAT (min)
250
225
200
175
150
125
100
75
50
EV
ICE

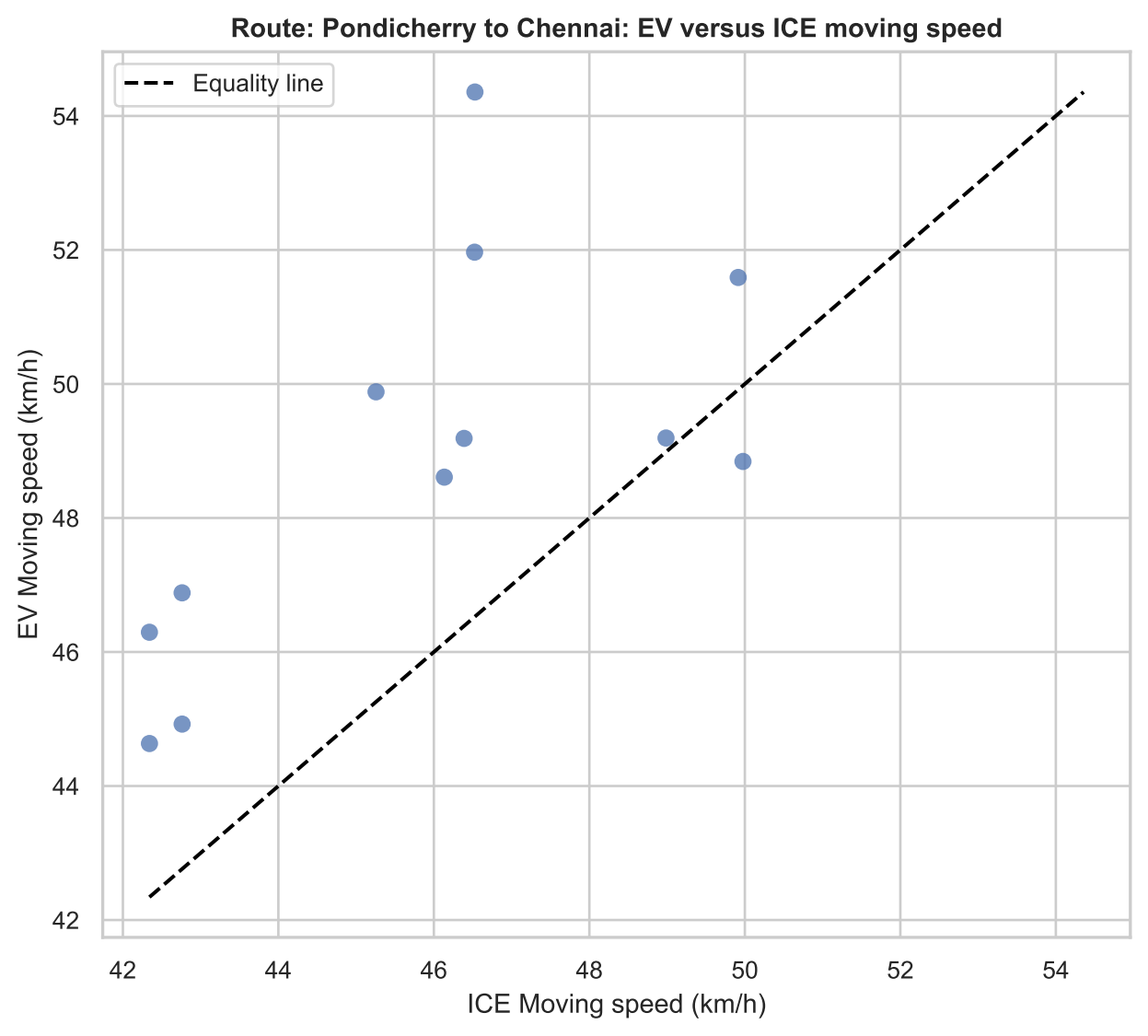

Route: Pondicherry to Chennai: Pair-by-phase time difference
pair_id
PAIR_022
PAIR_014
PAIR_011
PAIR_021
PAIR_023
PAIR_026
PAIR_015
PAIR_020
PAIR_029
PAIR_010
PAIR_012
PAIR_013
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
15
10
5
0
−5
−10
−15

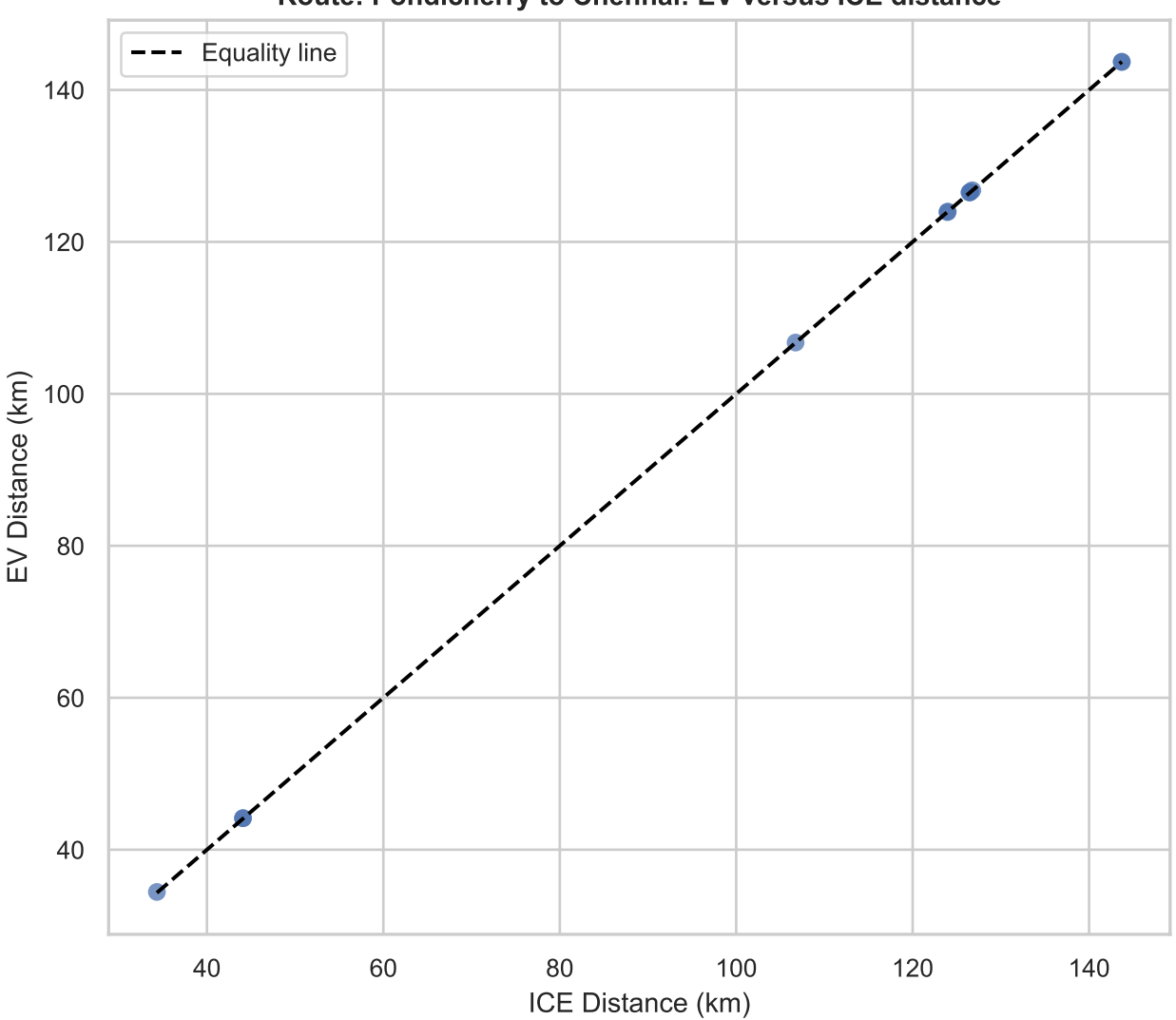

Route: Pondicherry to Chennai: Mean phase contribution to TAT difference
Mean EV – ICE phase time (min)
5
0
−5
−10
−15
−20
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

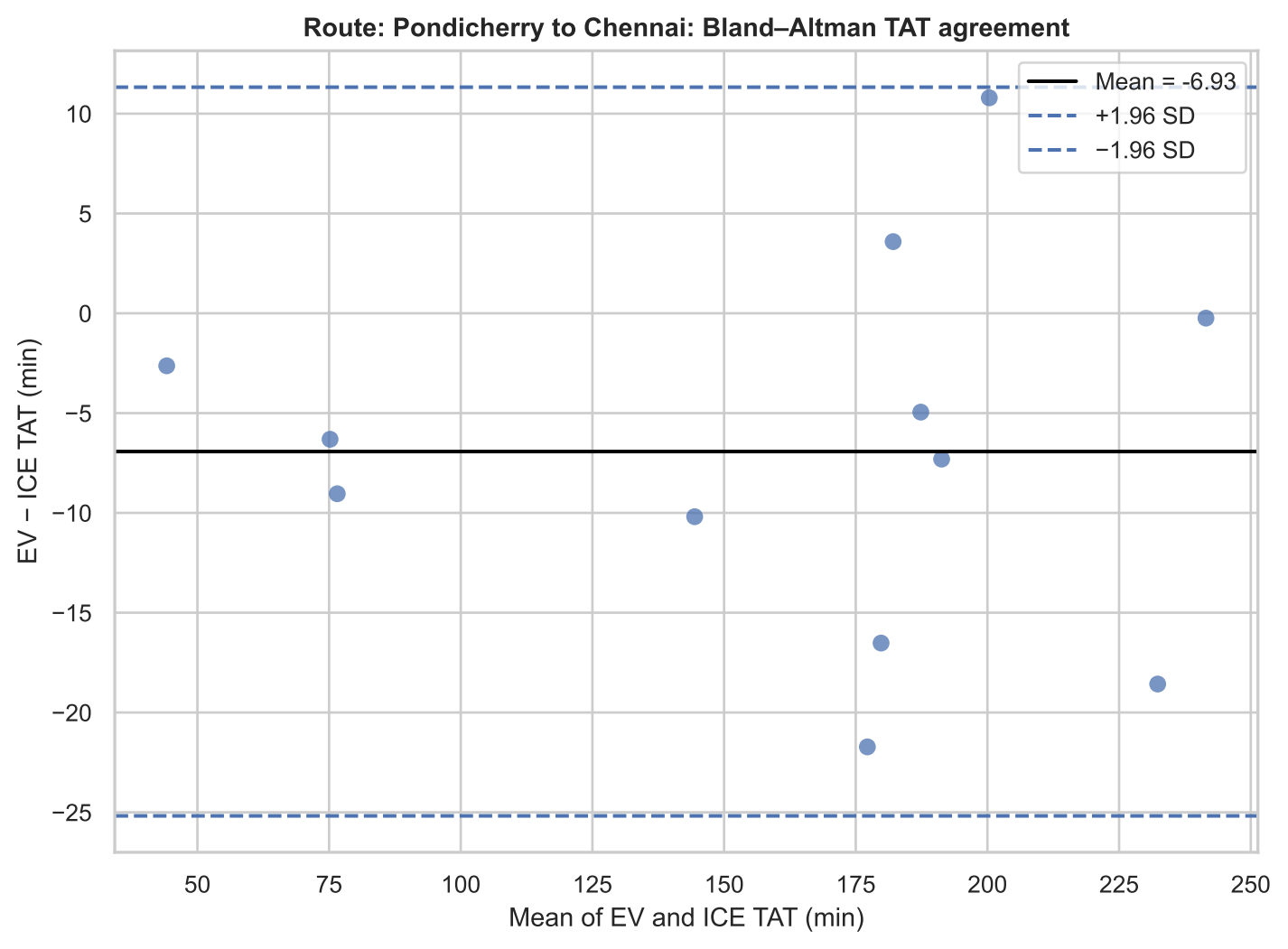

Route: Pondicherry to Chennai: Pearson correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00 -0.31 -0.66 0.58 0.34 0.73 0.40 0.38 0.41 0.15
-0.31 1.00 0.24 -0.58 -0.55 -0.18 -0.64 -0.34 0.23 0.68
-0.66 0.24 1.00 0.11 0.33 -0.54 -0.27 -0.74 -0.76 -0.09
0.58 -0.58 0.11 1.00 0.93 0.37 0.45 -0.14 -0.39 -0.26
0.34 -0.55 0.33 0.93 1.00 0.01 0.48 -0.23 -0.59 -0.38
0.73 -0.18 -0.54 0.37 0.01 1.00 0.01 0.19 0.46 0.27
0.40 -0.64 -0.27 0.45 0.48 0.01 1.00 0.51 -0.30 -0.59
0.38 -0.34 -0.74 -0.14 -0.23 0.19 0.51 1.00 0.46 -0.46
0.41 0.23 -0.76 -0.39 -0.59 0.46 -0.30 0.46 1.00 0.37
0.15 0.68 -0.09 -0.26 -0.38 0.27 -0.59 -0.46 0.37 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

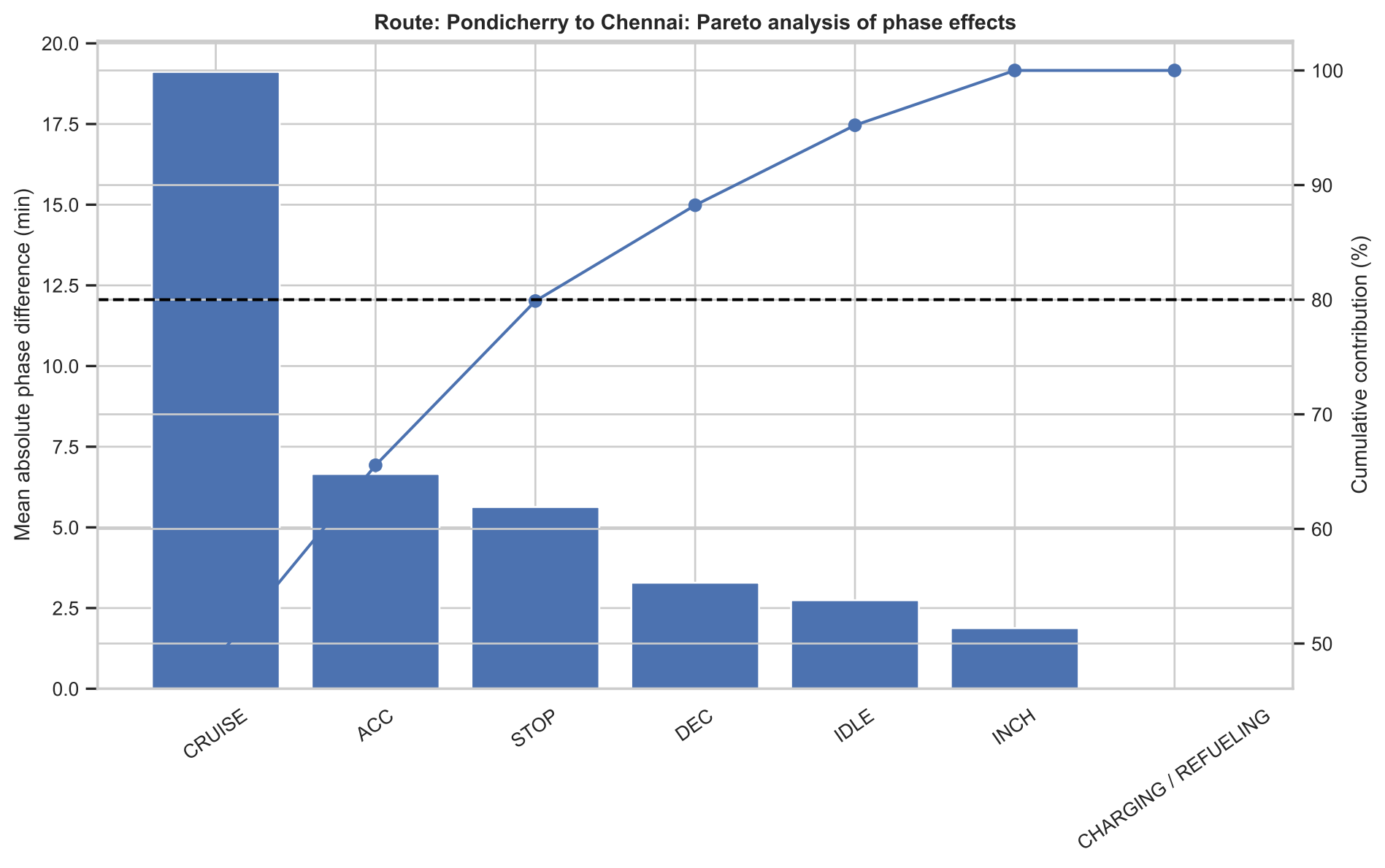

Route: Pondicherry to Chennai: Spearman correlation of pair-level differences
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00 -0.27 -0.59 0.43 0.17 0.73 0.33 0.35 0.43 0.08
-0.27 1.00 0.31 -0.54 -0.52 -0.12 -0.75 -0.53 0.25 0.80
-0.59 0.31 1.00 0.27 0.43 -0.57 -0.38 -0.68 -0.70 0.06
0.43 -0.54 0.27 1.00 0.90 0.25 0.24 -0.16 -0.45 -0.30
0.17 -0.52 0.43 0.90 1.00 -0.13 0.27 -0.23 -0.63 -0.36
0.73 -0.12 -0.57 0.25 -0.13 1.00 0.06 0.11 0.42 0.28
0.33 -0.75 -0.38 0.24 0.27 0.06 1.00 0.65 -0.18 -0.60
0.35 -0.53 -0.68 -0.16 -0.23 0.11 0.65 1.00 0.33 -0.63
0.43 0.25 -0.70 -0.45 -0.63 0.42 -0.18 0.33 1.00 0.28
0.08 0.80 0.06 -0.30 -0.36 0.28 -0.60 -0.63 0.28 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

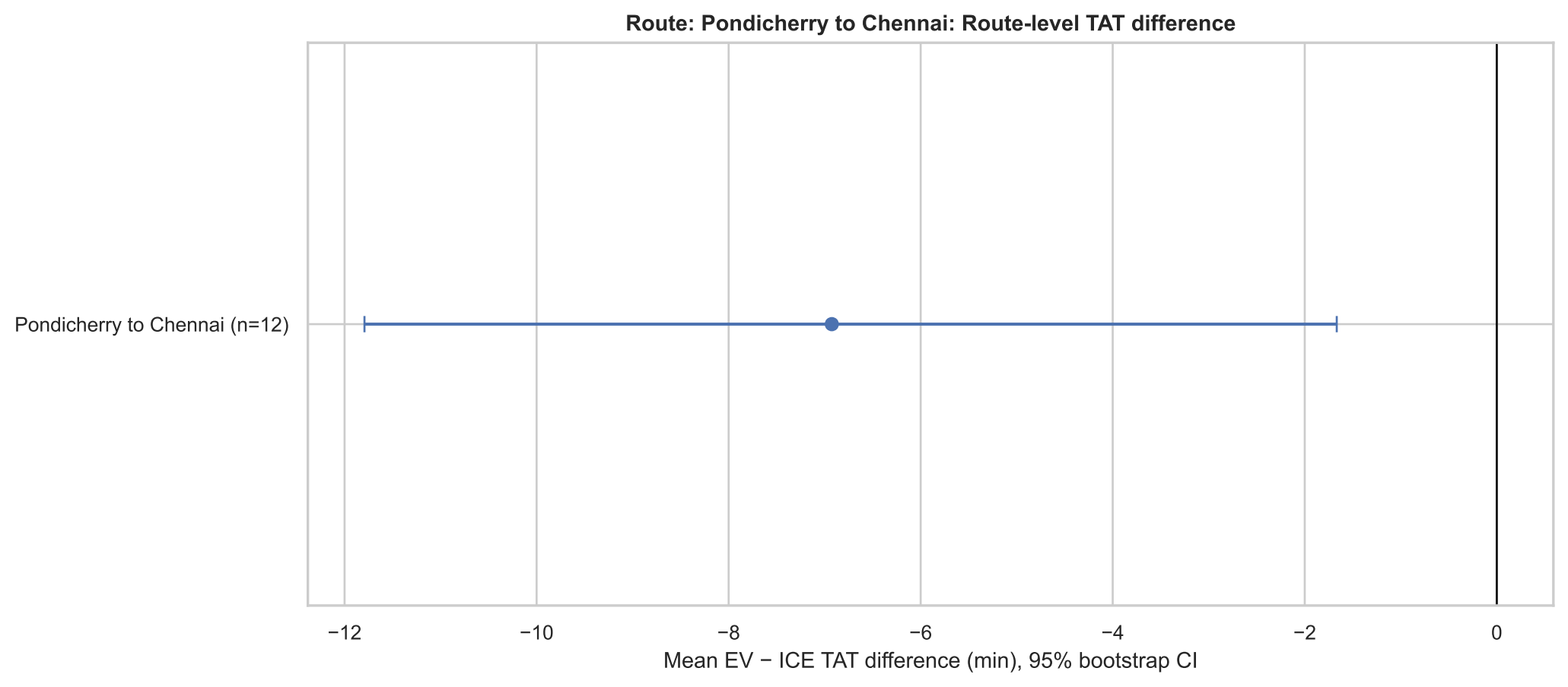

Route: Pondicherry to Chennai: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
250
225
200
175
150
125
100
75
50
50
75
100
125
150
175
200
225
250

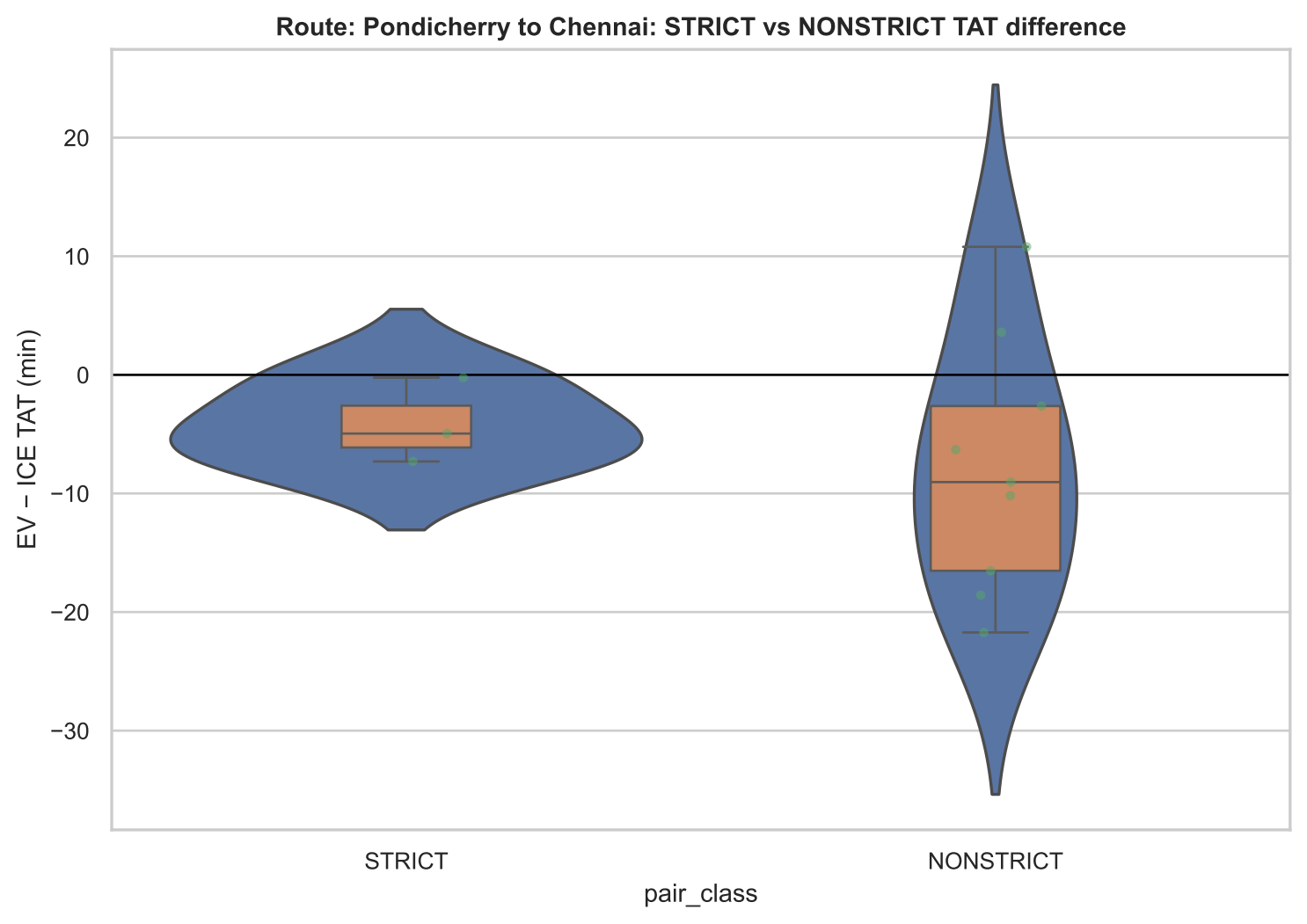

Route: Pondicherry to Chennai: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
42
44
46
48
50
52
54

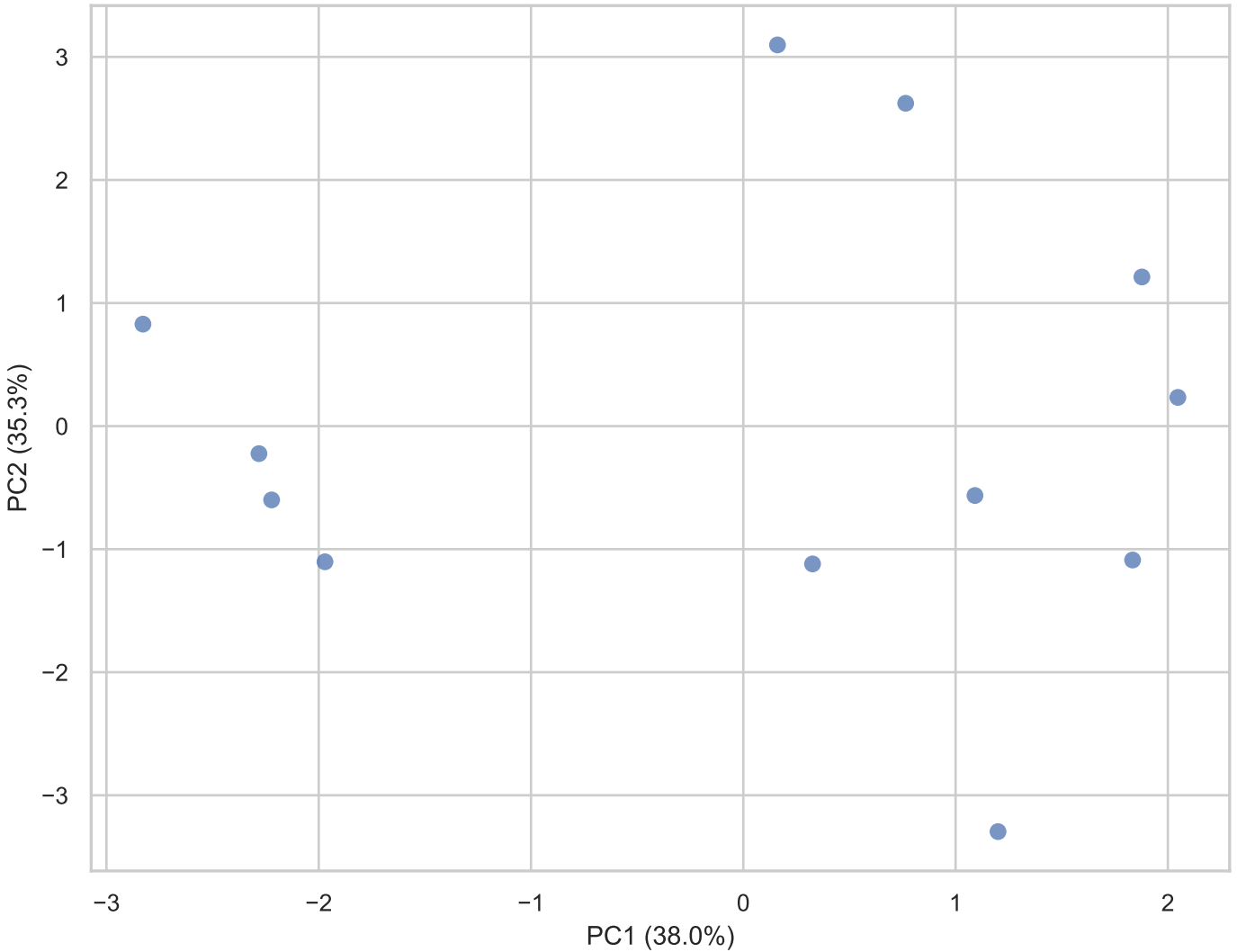

Route: Pondicherry to Chennai: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
40
60
80
100
120
140

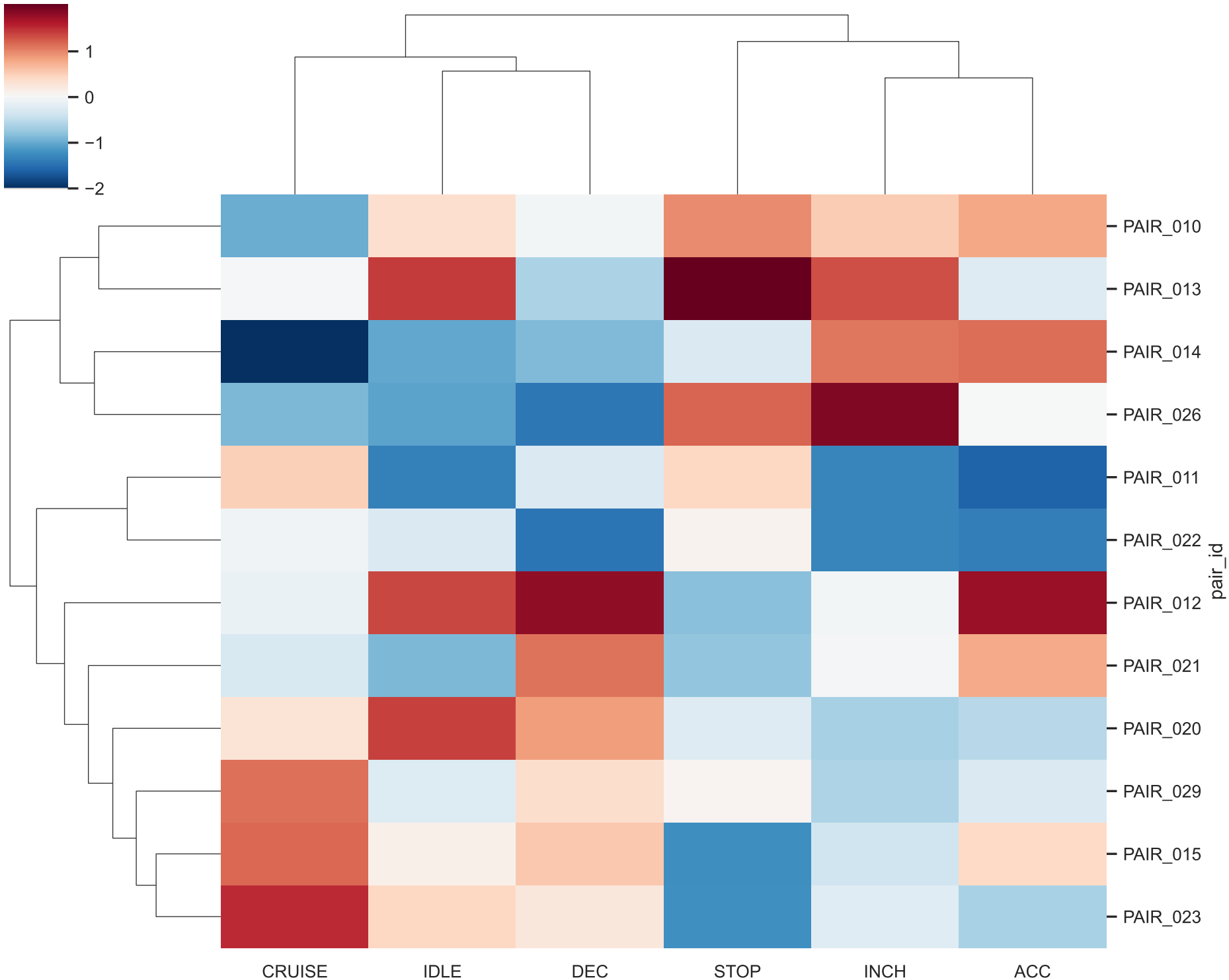

Route: Pondicherry to Chennai: Bland–Altman TAT agreement
Mean = -6.93
+1.96 SD
−1.96 SD
EV – ICE TAT (min)
Mean of EV and ICE TAT (min)
10
5
0
−5
−10
−15
−20
−25
50
75
100
125
150
175
200
225
250


Route: Pondicherry to Chennai: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
100
90
80
70
60
50
CRUISE
ACC
STOP
DEC
IDLE
INCH
CHARGING / REFUELING


Route: Pondicherry to Chennai: Route-level TAT difference
Pondicherry to Chennai (n=12)
−12
−10
−8
−6
−4
−2
0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


Route: Pondicherry to Chennai: STRICT vs NONSTRICT TAT difference
EV – ICE TAT (min)
20
10
0
−10
−20
−30
STRICT
NONSTRICT
pair_class


Route: Pondicherry to Chennai: PCA of EV−ICE operational differences
PC2 (35.3%)
PC1 (38.0%)
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2


Route: Pondicherry to Chennai: Hierarchical clustering of phase-difference signatures
1
0
−1
−2
PAIR_010
PAIR_013
PAIR_014
PAIR_026
PAIR_011
PAIR_022
PAIR_012
PAIR_021
PAIR_020
PAIR_029
PAIR_015
PAIR_023
pair_id
CRUISE
IDLE
DEC
STOP
INCH
ACC